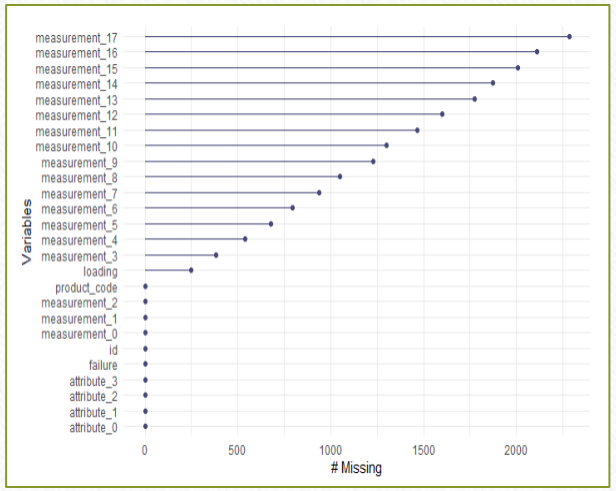
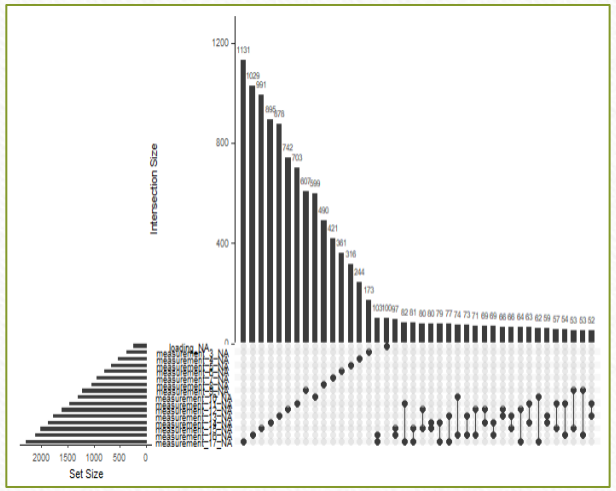
Product Manufacture Exploration

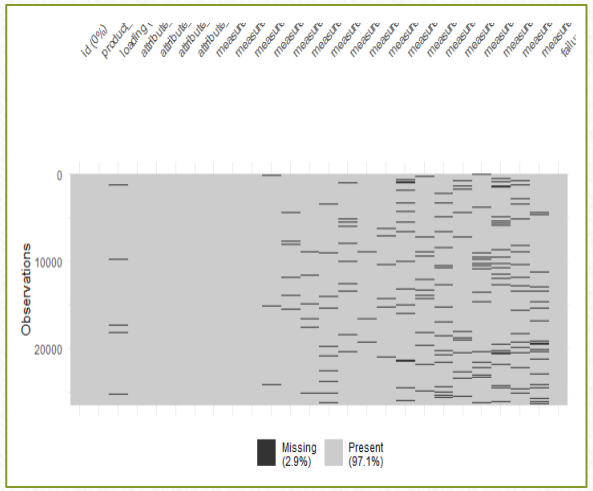
Developing Models Predicting Product Failure

Data Cleaning Strategy

Data cleaning is the preparation of raw data so it is accurate, consistent, and ready for analysis.



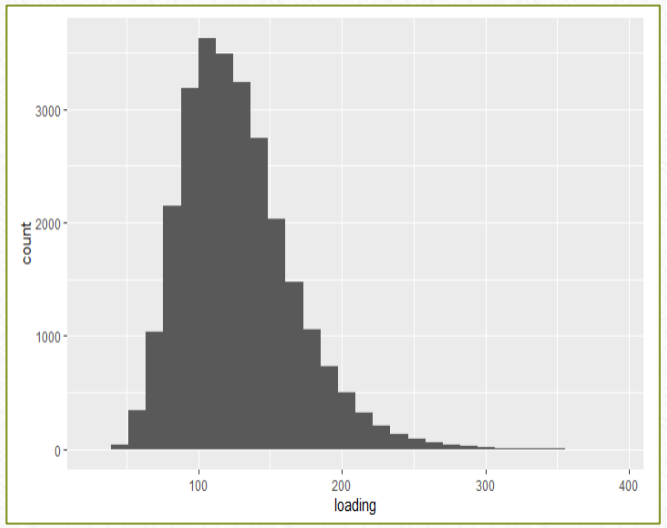




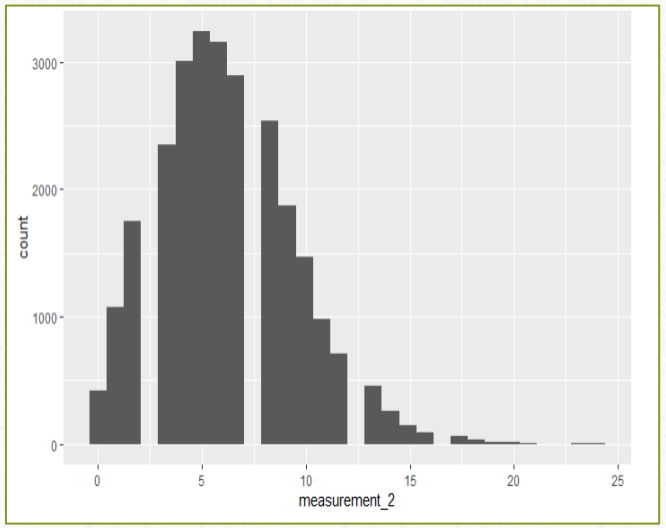
As an initial step, missing data was identified and explored for patterns of missingness. Some variables had a higher missingness rate. The missing data was determined to be Missing At Random (MAR).

Data Cleaning Strategy

Exploring data distributions for outliers in variables in the dataset



Example data distribution



Anomaly observed; intervals missing, outliers present

The distribution of each numeric variable was explored to pinpoint significant outliers and other anomalies. An example distribution of one of the dataset variables, "loading", is provided above. Outliers and variable distribution anomalies are noted for the "measurement_2" variable. This variable may be a strong predictor for product failure, as noted later. All other numeric variables showed reasonable distribution profiles. The outliers were filtered out for numeric variables where present, including "measurement_2" .

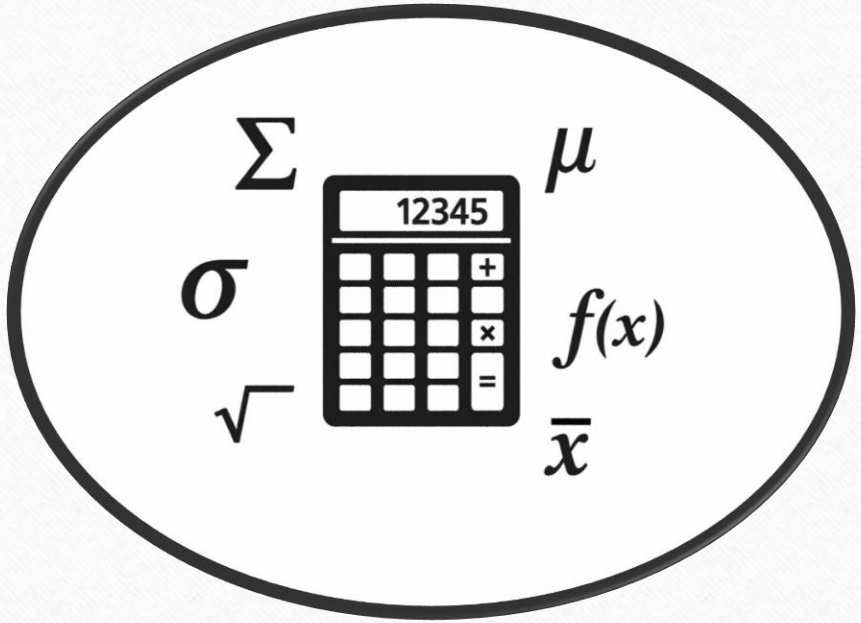
Data Cleaning Strategy

Missing data randomness verified, outliers deleted; data cleaned by imputation

Three ways to clean data:

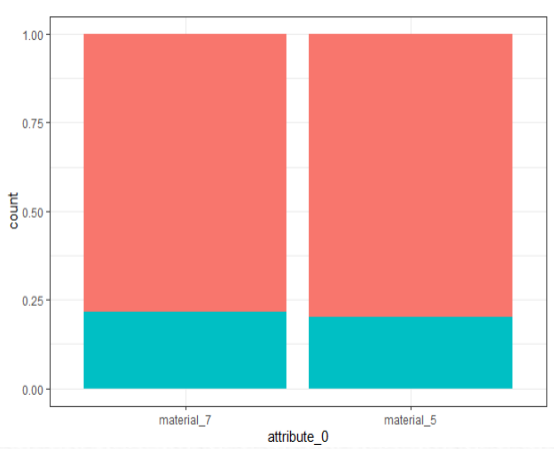
- Column-wise deletion
- Row-wise deletion
- Imputation

Dataset cleaned by row-wise deletion of outliers, then imputation: prepared/ imputed missing data with statistically formed values



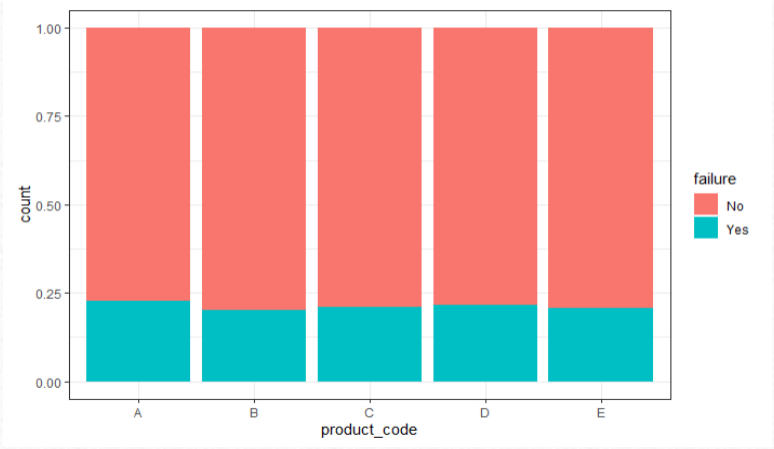
Variable Selection

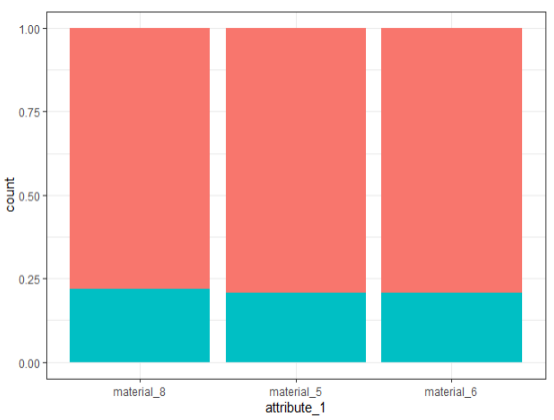
Identifying Strong Predictors of Failure: **Categorical Variables**



Attribute_0 Proportions

	Sample Size	Failures	Failure Rate
material_7	21320	4597	22%
material_5	5250	1052	20%





Attribute_1 Proportions

	Sample Size	Failures	Failure Rate
material_8	10865	2379	22%
material_5	10362	2164	21%
material_6	5343	1106	21%

Product_codeA is the strongest categorical variable predictor of failure

Product_code Proportions

	Sample Size	Failures	Failure Rate
A	5100	1159	23%
B	5250	1052	20%
C	5765	1220	21%
D	5112	1112	22%
E	5343	1106	21%

Variable Selection

Identifying Strong Predictors of Failure: **Numerical Variables**

Univariate Logistic Regression

Predictor Variable	P-Value	Estimate
measurement_17	4.07E-05	0.0006
measurement_5	6.80E-03	0.0461
attribute_3	1.29E-02	-0.0294
measurement_6	1.30E-02	0.0388
measurement_8	1.84E-02	0.0405

Binomial Logistic Regression

Predictor Variable	P-Value	Estimate
loading	4.12E-65	0.0072
product_codeB	1.21E-02	-0.2178
measurement_2	1.70E-02	0.0132

Small p-value: strong predictor of failure
(Significant p-value < 0.05)
Positive estimate: increases predictive odds of failure
Negative estimate: decreases predictive odds of failure

Summary

- Missing data and outliers were identified and explored
  - Outliers cleaned by row-wise deletion
  - Missing data cleaned by imputation
- Potentially strong predictors of product failure to utilize in future prediction models identified as:
  - Measurement_2,5,6, 8, and 17
  - Attribute_3
  - Loading
  - Product_codeA